Caterina | Visual Math Study Strategies

Five original, worked visual examples inspired by the numerical reasoning and word-problem formats on the two Wonderlic GAIN preparation websites. These are not copied test questions and are not official Wonderlic items.

How to use each page: read the question, stop and identify what is being asked, cover the answer, try one step, then uncover the worked explanation. Say the steps aloud if helpful.

1. Make fractions visible

Try it: A class has 12 practice questions. Caterina answers 9 correctly. What fraction and percent did she get right?

9 of 12 squares shaded



9 / 12 = 3 / 4 = 75%

Worked answer

Step 1: Put the number correct on top: 9/12.

Step 2: Simplify by dividing top and bottom by 3: 3/4.

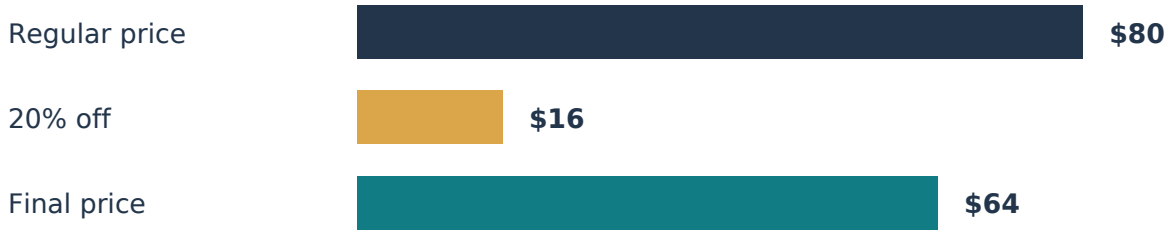
Step 3: Three of four equal parts = 75%. Answer: 75%.

WHY this strategy helps: The shaded blocks make the part-to-whole relationship visible, reducing the need to hold multiple numbers in working memory.

Stop-and-check: Does the percentage fit the picture? Nine shaded blocks is more than half but less than all twelve.

2. Translate a percent word problem

Try it: A $80 study kit is discounted by 20%. What is the final price?



Worked answer

Step 1: Underline what is asked: final price, not just the discount.

Step 2: Find the discount: 20% of $80 = 0.20 × 80 = $16.

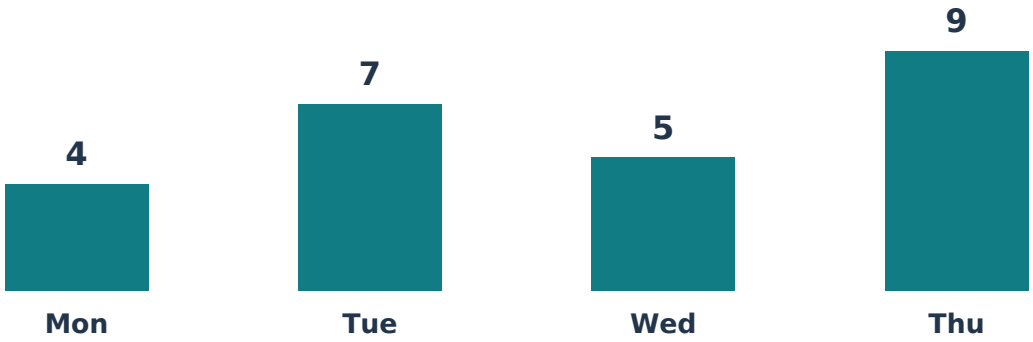
Step 3: Subtract: $80 − $16 = $64 final price.

WHY this strategy helps: Separating “discount amount” from “final price” helps prevent answering the wrong question, especially when instructions contain several numbers.

Stop-and-check: Is the final price lower than the original? Yes: $64 is less than $80.

3. Read a graph: find the difference

Try it: The bars show practice sets completed each day. How many MORE sets were completed Thursday than Monday?



Worked answer

Step 1: Locate the two days named: Thursday = 9; Monday = 4.

Step 2: “How many more” asks for a difference.

Step 3: Subtract: 9 − 4 = 5 more sets.

WHY this strategy helps: Visually locating only the requested bars filters out distracting information and links comparison language to subtraction.

Stop-and-check: Did you compare Thursday and Monday, rather than add every bar?

4. Compare quantities one side at a time

Try it: Which is greater: 8 × 6 or 120 ÷ 5?

Left side	Right side
8 × 6 = 48	**120 ÷ 5 = 24**

Worked answer

Step 1: Calculate left side: 8 × 6 = 48.

Step 2: Calculate right side: 120 ÷ 5 = 24.

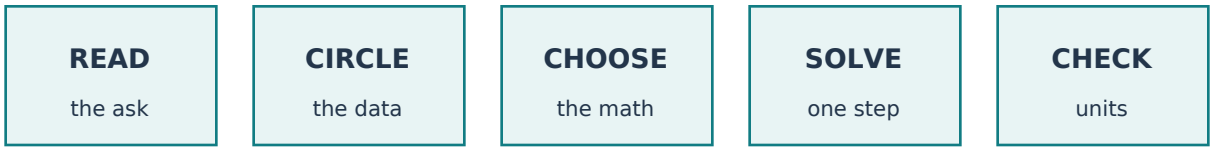
Step 3: Compare: 48 > 24. The left side is greater.

WHY this strategy helps: Two separate boxes externalize the intermediate answers, so the comparison does not depend on remembering both calculations at once.

Stop-and-check: Check the comparison symbol: the open side faces the larger number.

5. Use the same five-box routine

Try it: A student completes 6 practice sets each day for 4 days. How many sets total?



Worked answer

READ: Read the question once.

CIRCLE: 6 sets per day; 4 days; total sets.

CHOOSE: Repeated equal groups means multiplication.

SOLVE: 6 × 4 = 24 sets.

CHECK: 6 + 6 + 6 + 6 = 24. The units are sets.

WHY this strategy helps: A repeatable visual routine builds a deliberate pause before action and reduces the executive-function burden of deciding how to begin every new problem.

Stop-and-check: Can she name the question and the units before calculating?

Coach / tutor notes

Begin with a single example, not a timed page. Ask Caterina to point to the information she needs and explain the first step. Allow silent processing time before prompting. When she is ready, cover the worked answer and repeat with different numbers.

The concussion history does not establish ongoing symptoms. Adjust session length or sensory demands according to her current experience, not an assumption.

Sources and scope

Practice question styles were informed by AssessmentCentreHQ, “Wonderlic GAIN Test” (assessmentcentrehq.com/wonderlic-gain-test/) and Practice4Me, “Wonderlic GAIN Test” (practice4me.com/wonderlic-gain-test/). The five questions, diagrams, worked answers and study explanations in this guide are newly authored. Caterina’s previously shared GAIN report informed the selection of fractions, percentages, graphs and word problems.